

PREDICTING WILDFIRE INTENSITY

Predicting Wildfire Intensity and Risk using Environmental Indicators in Seven Fire-Prone
Regions Globally

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Introduction

Wildfires are undergoing a massive transformation as greenhouse gas (GHG) emissions intensify global warming, creating the types of conditions that increase the frequency and duration of fire type events. Fire seasons have been extended globally and forest-fire carbon emissions have risen approximately 60% from 2001 to 2023 with extreme wildfire activity more than doubling in many regions (Nasa. 2025). The fallout from wildfire occurrences ranges from displacement of local habitats and their wildlife, all the way to large fatality counts amongst highly affected populations near or within wildfire areas.

The research conducted herein will examine if environmental and satellite-derived data can classify wildfire fire intensity across seven global regions. Additionally, the research will determine if the environmental and satellite-data is enough to develop predictive capacity for early warning systems of extreme wildfires in an effort to improve upon fire management and resource allocation needs. We will use models such as logistic regression, random forests, clustering algorithms such as k-Nearest Neighbors to understand the problems and predictive possibilities of wildfire risks.

Methodology

The objective of this study is to determine if environmental and satellite-derived data are sufficient to both classify and predict wildfire intensity in seven global regions. The intent is to use this model to predict the intensity of future wildfires based on the indicators in this historical dataset.

Dataset Description

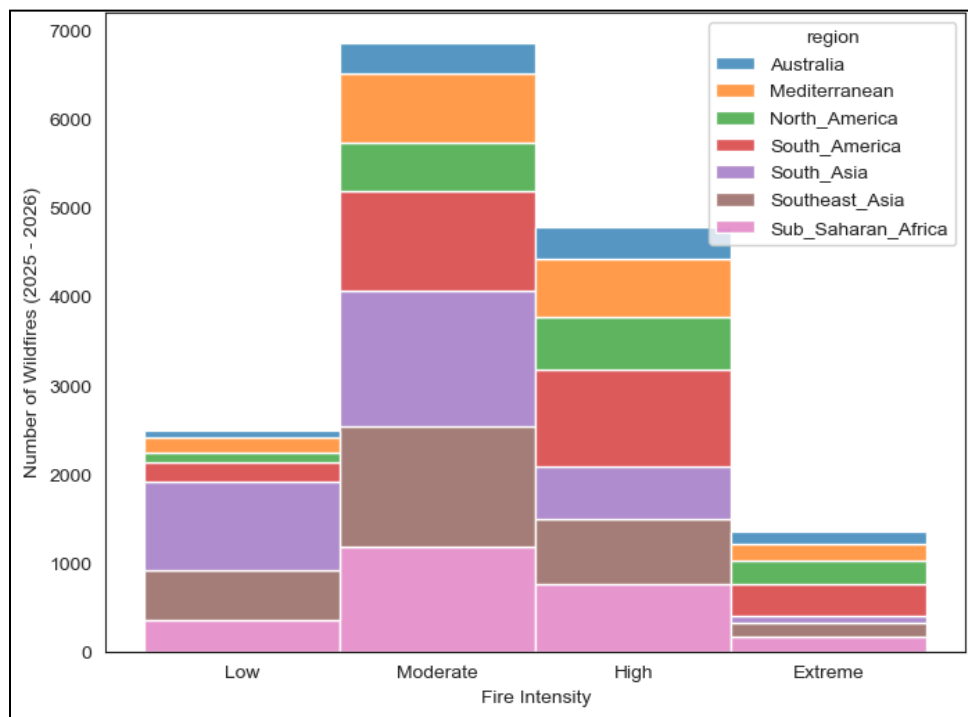
The dataset that will be used documents wildfire data collected from satellites in 7 fire-prone regions globally in South Asia, Southeast Asia, Mediterranean, Sub-Saharan Africa, South America, North America, and Australia. A total of 15,500 fire detection events across 35 countries are included

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covering January 2024 – December 2025. Thirteen variables are included for each event, covering fire intensity indicators such as temperature, radiative power, and fuel type as well as weather indicators such as air temperature, wind speed, precipitation, and humidity.

The dataset has a pre-existing categorical classification engineered for wildfire intensity for 4 classes of low, moderate, high, extreme. This existing categorical classification (fire_intensity) is retained as the target variable. Figure 1 shows the fire intensity by world region. Class rebalancing was considered for the dataset given the extreme wildfire classification constituted 9% of the total, however rebalancing was implemented within each model instead of holistically in the preprocessing phases.

Figure 1 - Fire Intensity Count by Region



Data Preprocessing

Columns were assessed for relevance, redundancy, and multicollinearity when determining the final features to include in the model. The exclusion of highly correlated features causing multicollinearity will be determined during Exploratory Data Analysis (EDA) and thus removed from the

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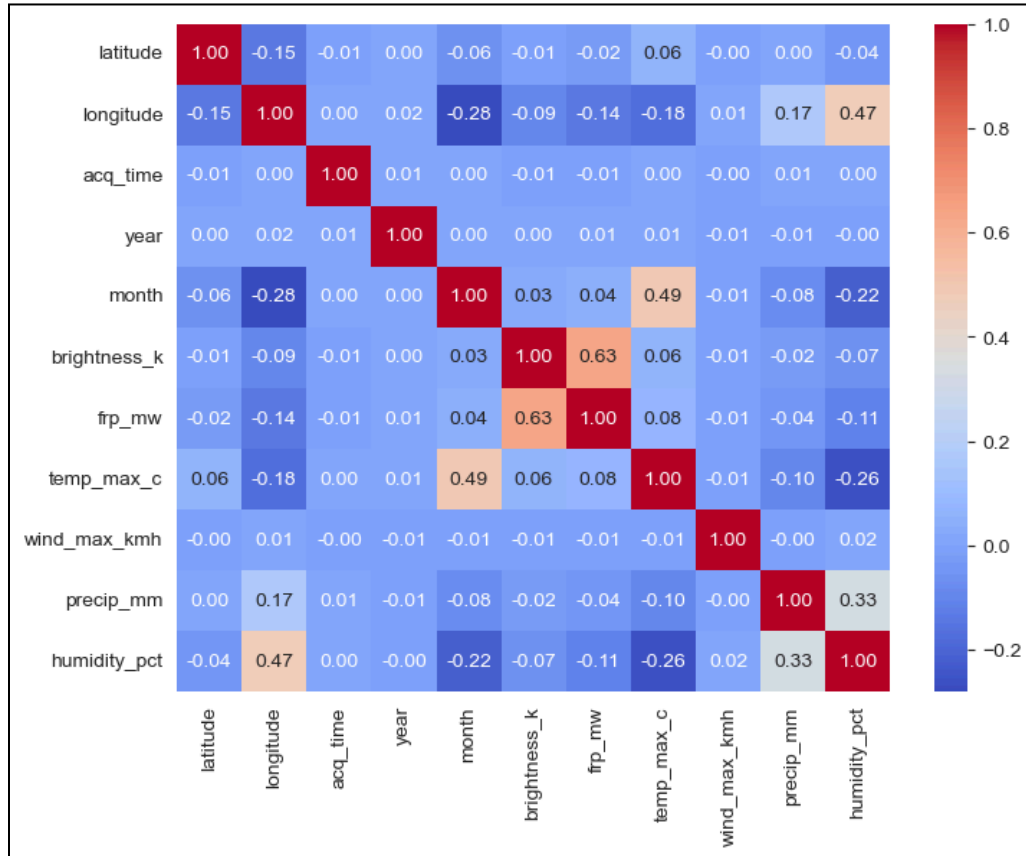
machine learning models that are used within this process to reduce the computational cost and time of modeling. When sending these models to operational teams for entry into fully scaled systems it is paramount for the data practitioners to account for multicollinearity and redundancy to enhance the model's timeliness and output.

Data about the satellite and instrument that captured the data were excluded as they are metadata and not relevant to this study. Latitude, longitude, and country were excluded due to redundancy with *region* as well as with more descriptive continuous variables such as temperature and humidity. Time of data collection, including *acq_date*, *acq_time*, *month*, *year*, and *daynight*. Confidence is not directly relevant to the model, but was retained to help troubleshoot models if necessary. Variables such as max air temperature on detection day in Celsius (*temp_max_C*), max wind speed in kilometers per hour (*wind_max_kmh*), precipitation (*precip_mm*), land type (*fire_type*), and percent humidity (*humidity_pct*) were retained for models aiming to predict conditions for more extreme fires. Categorical variables from the dataset were encoded to allow for processing within classification and predictive modeling.

All numerical variables were assessed for multicollinearity. The features *frp_mw* and *brightness_k* are both correlated at the highest levels (Pearson Correlation Coefficient = 0.63). While this correlation coefficient does not indicate problematic multicollinearity, it does flag the need for further research into the cause for a higher correlation value. Both *frp_mw* and *brightness_k* are satellite measurements of the fire energy and temperature when burning, so a higher correlation makes intuitive sense. The remainder of the numerical values were deemed not multicollinear and can be seen in Figure 2.

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Figure 2 - Correlational Heatmap (Pearson's Correlation)



Data Leakage from Fire Radiative Power and Brightness

Fire radiative power (*frp_mw*) and brightness (*brightness_k*) are satellite measurements of the energy and temperature of a burning fire rather than a predictor of it. Fire radiative power measures the rate at which energy is radiated from a fire and brightness measures the brightness temperature in kelvin from the satellite pixel of the flame. When classification models use *frp_mw* and *brightness_k* as variables, the classification models run at almost perfect accuracy, and when it is taken away, models run at a far lower accuracy. Fire intensity is classified using Byram's Fire Intensity calculation, which calculates the rate of energy release per unit of time per unit of length (Alexander, 1982). Since fire radiative power is used directly in the binning of this classification scheme, it is a strong variable for a classification model. However, it is clear that *frp_mw* would not be an appropriate variable for a fire

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intensity prediction model as a future fire has no radiative power. As such, this study will assess both a classification model and prediction model separately. Figure 3 and 4 show the spread and dispersion of *frp_mw* and *brightness_k* across intensity classifications.

Figure 3 - Fire Radiative Power versus Fire Intensity

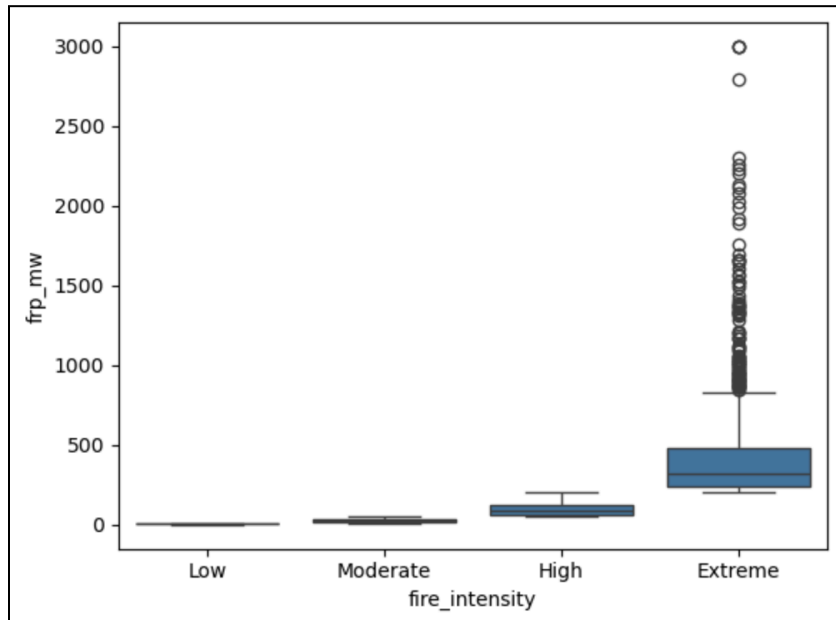
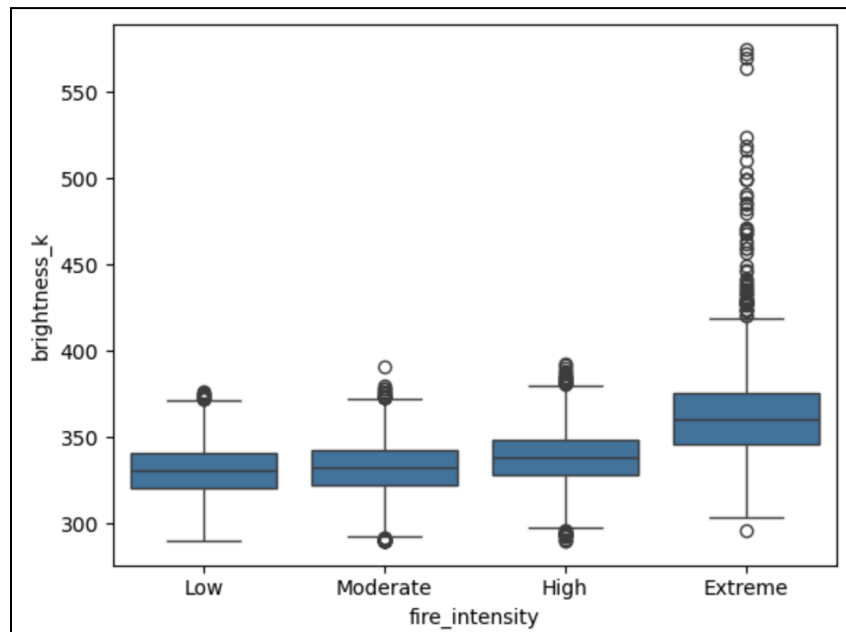


Figure 4 - Fire Brightness versus Fire Intensity



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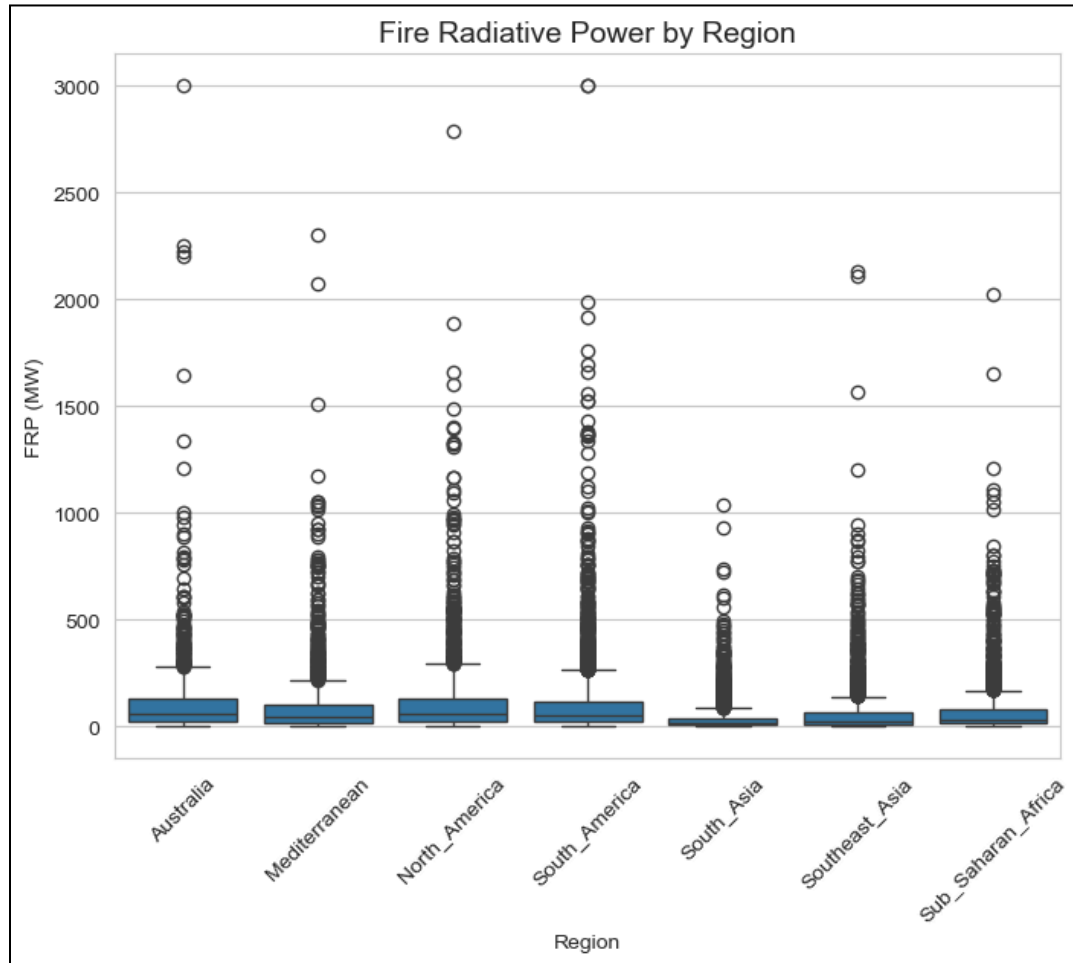
Multiple Model Assessment:

Researchers tested multiple combinations of features within the Exploratory Data Analysis (EDA) portion of this research to conclude important topics of note that can be verified to be predictive in nature. During the EDA, researchers concluded that Fire Radiative Power by Region (FRP_MW by Region - Figure 5) varied greatly and that the Mediterranean, Southeast Asia, Sub-Saharan Africa, and, lastly, South Asia came in with the lowest Radiative Power in all regions. The highest regions were Australia and South America (approximately 3000) with North America coming in third amongst the leaders. More research will have to be conducted that is out of the scope of the dataset contained within this research paper to examine wind patterns that flow through various latitudinal and longitudinal lines that could be contributory to wildfire occurrences in relation to the Greenhouse Gas variations in the highest regions of FRP_MW. Other contributory factors such as precipitation, humidity, wind speed, etc. could be taken into account when dealing with directional wind flows of the unknown type.

We will take into account all of the metrics when evaluating the final output of all of the models, but understand the importance of the F1 Score and Recall when dealing with wildfire risks. A higher overall average of both precision and recall can be advantageous due to the considerations met by both metrics in evaluations. All regions were included in the classification and modeling explorations, and there is major consideration for a single region model versus a multi-region model. Both will be explored.

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Figure 5 - Fire Radiative Power by Region



Machine Learning Models

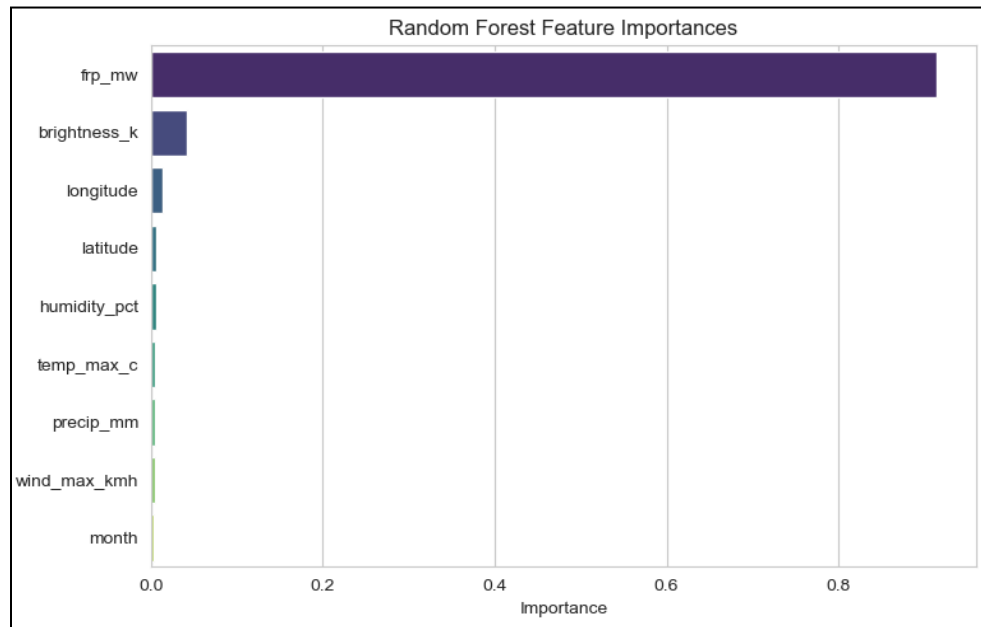
To test and validate these machine learning classification models for the 13 features within the dataset, which held a maxima of 15,500 data points, we tested multiple classification models including Random Forest, k-Nearest Neighbor, and Logistic Regression and aggregated their results here within. A Naive Bayes model was considered but not used as the model assumption of including only independent variables was violated since weather metrics are not independent. A neural network model was also considered but not included as it was ultimately deemed to be too complex for this type of classification and prediction scenario.

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Random Forest

A Random Forest is an ensemble type of machine learning model that creates recursive decision trees to compare data. These decision trees within the Random Forest will output the best possible split or important feature within a dataset via a set of given directives. The computational cost of the Random Forest is $O(n)$ complexity (where n represents the number of samples) when dealing with large amounts of data. A Random Forest model was used to observe if there would be any naturally predicted variables of note that could be valuable during the full modeling process. A maximum value three depth tree (3) with the GINI inequality was used to determine splitting, and, as noted in EDA, the feature - Fire Radiative Power (FRP_MW) was the Random Forest's natural choice for creating decision trees. Figure 6 is a descending barplot of tree output features based upon the features' importance.

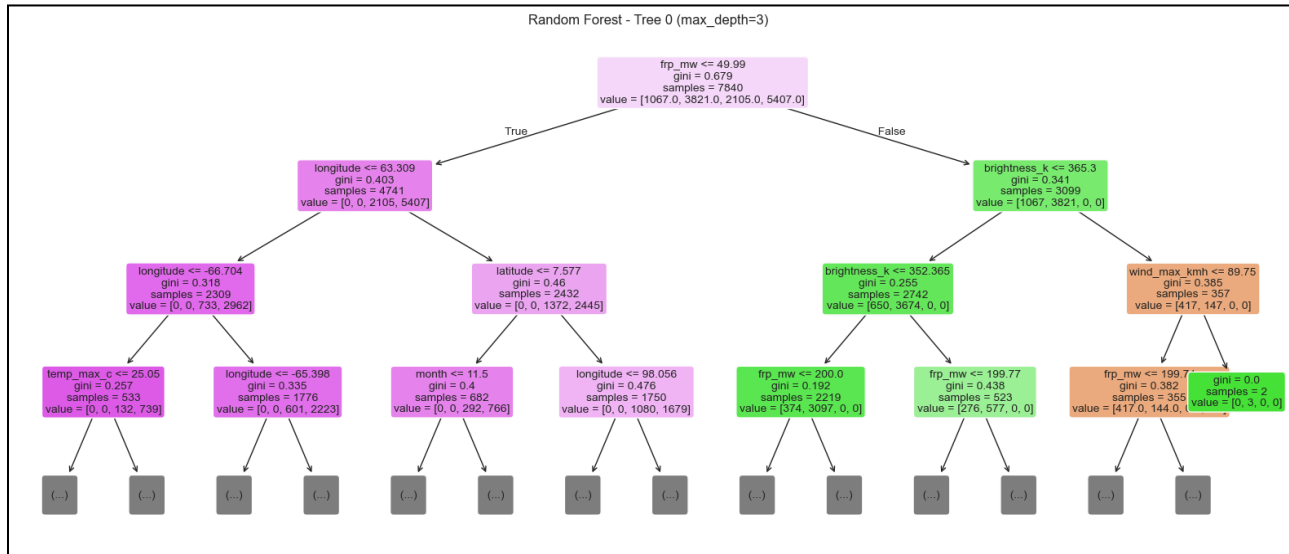
Figure 6 - Random Forest Feature Importances



The best decision tree (Figure 7) contained splits amongst the most important values FRP_MW, brightness_k, and longitude.

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Figure 7 - Random Forest - Decision Tree (FRP_MW)



The total predictable power of the Random Forest would predict the most important feature of fire prediction (FRP_MW) at 100% Accuracy. The precision, recall, and f1-score also were predicted at 100%. More than likely is that the Fire Radiative Power is the outcome of wildfire intensity.

Although it is not beneficial for prediction to take place here using this feature, it is important to understand these nuances during the data mining process. Understanding of topics of this nature will ensure that research is smooth and that data values are true in their classification and predictive nature. Data practitioners should continue to be vigilant for metrics of higher than 98% as this could imply overfitting within the testing phases, and possibly the assumption of noise within testing as normal. Models such as this example will not generalize well in true predictive forecasting.

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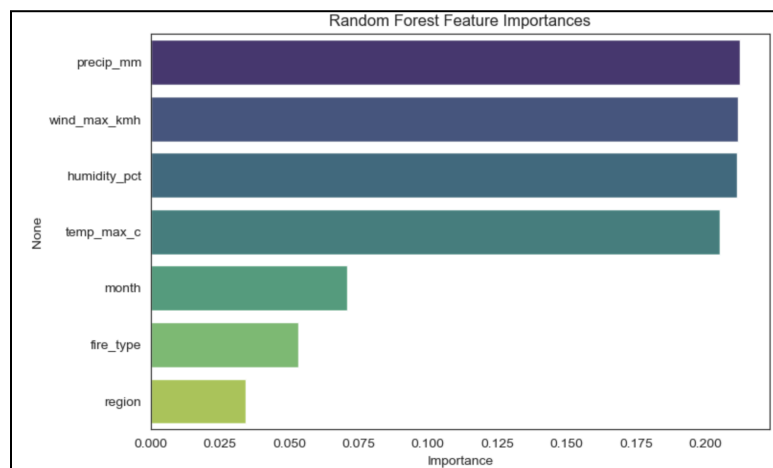
Accuracy: 0.9996774193548387				
Classification Report:				
	precision	recall	f1-score	support
Extreme	1.00	1.00	1.00	272
High	1.00	1.00	1.00	957
Low	1.00	1.00	1.00	499
Moderate	1.00	1.00	1.00	1372
accuracy			1.00	3100
macro avg	1.00	1.00	1.00	3100
weighted avg	1.00	1.00	1.00	3100
Confusion Matrix:				
[[272 0 0 0]				
[0 957 0 0]				
[0 0 498 1]				
[0 0 0 1372]]				

Table 1 - Random Forest Classification Metrics with Confusion Matrix

Random Forest for Prediction

A Random Forest was re-run without the *frp_mw* and *brightness_k* features, instead relying only on environmental factors that may increase the danger of a higher intensity fire. As shown in Figure 8, relative importance of weather-based indicators have increased. This model has reduced accuracy, with a weighted average of 43% accuracy in predicting the intensity of fires. The value of importance within features has decreased from the top of *frp_mw* (Importance = 0.91) to the top of *precip_mm* (Importance = 0.22).

Figure 8 - Random Forest Model



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The table below (Table 2) shows the metrics surrounding the model *not* using the most important feature of the dataset ‘frp_mw’ in the modeling. The accuracy fell to .43, the ‘Extreme’ precision to .21 from 1.0, the recall decreased to .03 from 1.0, and, lastly, the F1 score fell to .05 from 1.0 - this shows significant loss of accuracy and possibly points to causal reasoning for Fire Intensity due to Fire Radiative Power. More research would be necessary to validate this conclusion.

Accuracy: 0.42548387096774193				
Classification Report:				
	precision	recall	f1-score	support
Extreme	0.21	0.03	0.05	271
High	0.37	0.32	0.34	967
Low	0.27	0.09	0.14	470
Moderate	0.46	0.69	0.55	1392
accuracy			0.43	3100
macro avg	0.33	0.28	0.27	3100
weighted avg	0.38	0.43	0.38	3100

Table 2 - Random Forest Model Metrics

K-Nearest Neighbors (K-NN)

The K-Nearest Neighbors model is a distance-based approach to machine learning that determines the distance between points and assumes similarity. The smaller the distance, the more the similarity. So, in relation to the data points within the wildfire risk dataset, these points were of medium to high accuracy.

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Figure 9 - K-Nearest Neighbor Model

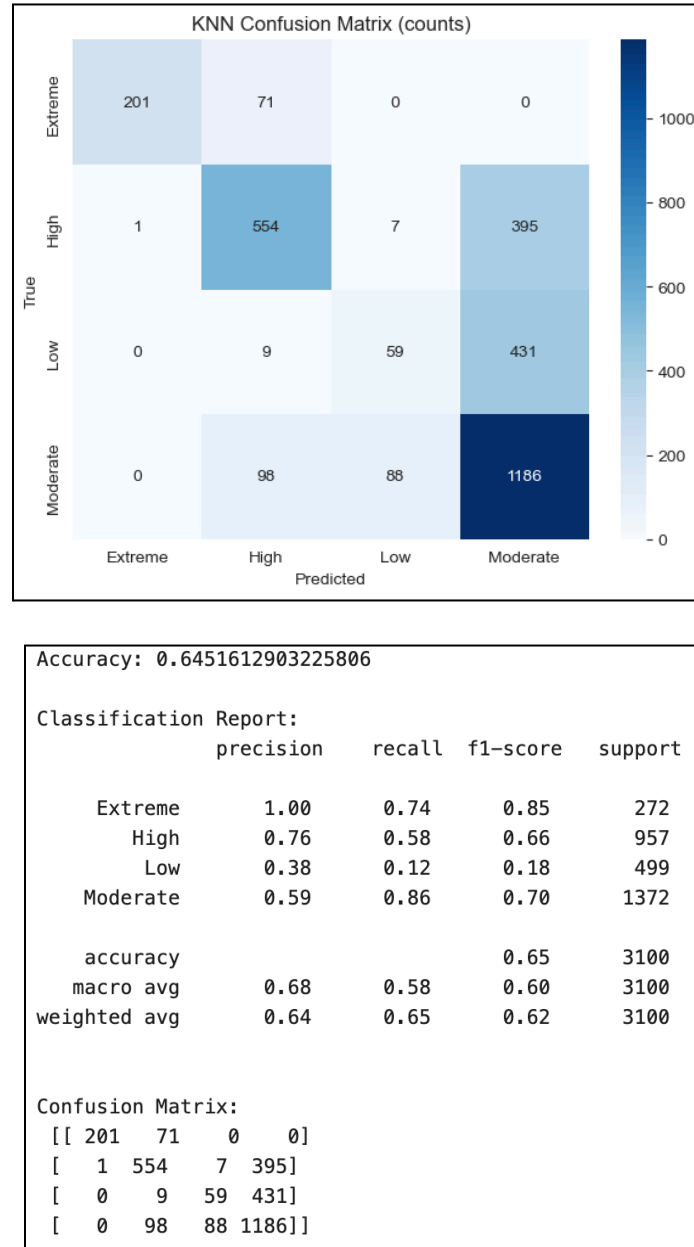


Table 3 - K-Nearest Neighbor Model Metrics

Logistic Regression

Logistic Regression models are either binary or multi-class discriminative models that predict outcomes of either yes or no, true or false, '1' or '0'; that is some form of outcome by way of the odds estimations of a data instance (x) using its attribute values (Tan et al. 2019). The Logistic Regression

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model determined the binary outcome of fires being predicted using relevant features, but not Fire Radiative Power (due to it completely having the predictive power to determine 100% accuracy). The Logistic model predicted values with an accuracy of 0.55.

Accuracy: 0.5451612903225806				
Classification Report:				
	precision	recall	f1-score	support
Not Evident	0.21	0.66	0.32	499
Evident	0.89	0.52	0.66	2601
accuracy			0.55	3100
macro avg	0.55	0.59	0.49	3100
weighted avg	0.78	0.55	0.60	3100
Confusion Matrix:				
[[329 170]				
[1240 1361]]				

Table 4 - Logistic Regression Metrics Report with Confusion Matrix

Results

Predictive Modeling

Machine Learning models are compared by many different metric categories, but their metrics like Accuracy, Precision, Recall, and F1 Score, typically, are good indicators of performance.

Accuracy

The results of the Predictive Modeling point towards the K-Nearest Neighbor (K-NN) as the highest or most accurate predictors for wildfire risk at a .65 Accuracy level compared to the Logistic Regression at .55 Accuracy and the Random Forest Predictor at .43. Accuracy is a measure of correct predictions to total predictions as outlined by algorithm 1.

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$$Acc = \frac{True\ Positives\ (TP) + True\ Negatives\ (TN)}{True\ Positives\ (TP) + True\ Negatives\ (TN) + False\ Positives\ (FP) + False\ Negatives\ (FN)} \quad (1)$$

Precision

The K-NN proves to not be a better predictor when dealing with the precision in regards to the other models used within this research paper. The precision metric measures the amount of accuracy of true positives called the true positive ratio that is gathered using the following algorithm (2).

$$Precision = \frac{True\ Positives\ (TP)}{True\ Positives\ (TP) + False\ Positives\ (FP)} \quad (2)$$

The K-NN precision score was a weighted average of .64 compared to that of the Logistic Regression and Random Forest, respectively, at .78 and .38. This determines that the precision of the Logistic Regression scored a higher true positive ratio than the other models within.

Recall

The recall score equates the ability for a model to predict all relevant positive instances within a dataset score. The below algorithm outlines the recall score.

$$Recall = \frac{True\ Positives\ (TP)}{True\ Positives\ (TP) + False\ Negatives\ (FN)} \quad (3)$$

The results of the recall score points towards the K-Nearest Neighbor (K-NN) as the highest at a .65 recall level compared to the Logistic Regression at .55 Accuracy and the Random Forest Predictor at .43.

F1 Score

Lastly, the F1 Score is the model's ability to score the harmonic mean of the precision and recall scores, and is a good way to measure imbalanced class datasets. This measure creates a balance among data

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points and is robust against noise and outliers. It will balance the need for high precision and high recall.

The algorithm (4) below outlines the F1 Score.

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

The K-NN algorithm received an F1 Score of .62, the Logistic Regression received a .60, and the Random Forest received a .38. The K-NN algorithm saw the highest harmonic mean of both precision and recall, but seemed to not be far ahead of the Logistic Regression model by much.

The prediction models could be improved if there were other factors within the dataset that were stronger influencers. Having a different model for different regions could possibly assist in the model's performance, but after creating a Logistic Regression using only the North American region there was a lower accuracy score resulting in the observation that a single region model might not perform as well using the same specific features which achieved a .49 accuracy.

We hypothesize that, in this predictive arena of wildfire risk, it is extremely important to predict the severity of wildfire due to these influential features using recall to reduce the amount of false negatives and that the metrics category of Recall might outweigh others. Only further research into the topic can determine if this hypothesis holds true or can be disproved.

Conclusion

Using the subject dataset, wildfire risk can be adequately predicted at a medium to medium-high level using one or multiple machine learning models to create an ensemble learning model that allows for the future predictability of low, medium, high, and extreme fire intensity. Using a severity based classification detects past fires extremely well and is perfectly accurate due to data leakage. A predictive

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model could certainly flag severe fire danger before a fire happens. If more informative feature data could be gathered such as prior environmental conditions (such as flagging drying conditions through historical temperature and precipitation data), would increase the predictive nature of these wildfire risk models.

Adding that additional data has the potential to decrease time spent trying to predict fires, preserve natural resources, and possibly save lives, which are the main goals of wildfire prediction.

References

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Extra Figures and Tables

Figure 10 - Fire Radiative Power $\times$ Brightness in Kelvin with Fire Type Clusters

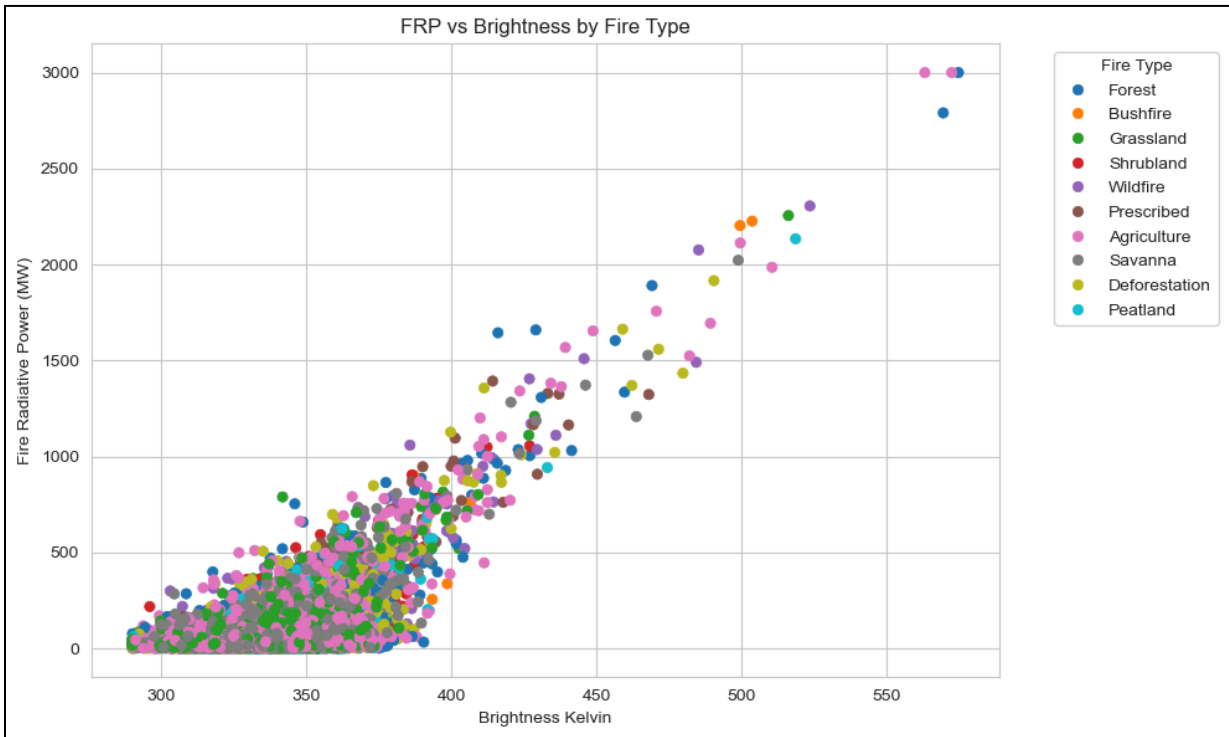


Figure 10 Zoomed - Largest Values of Brightness Kelvin (X) and Fire Radiative Power (Y)

